

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Integrated Experiments

Weightage: 40%

QUESTIONS:

Total Marks: 30

Experiment 1: Responsive Layout Design

Suppose that a retail website is not responsive across mobile and tablet devices, causing misalignment of content and improper spacing.

- (a) Analyze the possible reasons for lack of responsiveness in the existing layout.
- (b) Design a responsive webpage layout using **CSS Flexbox or Grid**, ensuring proper alignment of header, navigation, content, and footer.
- (c) Demonstrate how the layout adapts to **different screen sizes (mobile, tablet, desktop)** using media queries.
- (d) Implement spacing, alignment, and distribution techniques to improve visual consistency.
- (e) Evaluate the effectiveness of the responsive design and suggest further improvements.

Experiment 2: Form Validation using JavaScript

Suppose that a registration form accepts invalid email and phone number inputs, leading to incorrect data submission.

- (a) Analyze the issues in the existing form validation mechanism.
- (b) Design input fields for email and phone number with appropriate HTML attributes.
- (c) Implement **JavaScript validation using Regular Expressions** for both email and phone number.
- (d) Develop a mechanism to display **real-time error messages** for invalid inputs.
- (e) Evaluate the validation system and suggest enhancements for better user experience and security.

Experiment 3: CSS Layout Debugging

Suppose that a webpage layout overlaps when the browser window is resized, affecting usability.

- Analyze the CSS properties responsible for layout overlapping issues.
- Examine the role of the **CSS box model (margin, border, padding, content)** in layout design.
- Debug the layout using appropriate **positioning techniques (relative, absolute, flex, grid)**.
- Modify the CSS to ensure proper alignment and prevent overlapping during resizing.
- Evaluate the corrected layout and suggest best practices for maintaining consistency.

Code of Conduct:

I Deepak V-192411021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
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Experiment 1: Responsive Layout Design

Code:

<!-- Experiment 1 - Responsive Layout Design -->

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Responsive Layout Design</title>

<style>

```
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial,sans-serif;
}
```

```
body{
background:#f4f4f4;
}
```

```
header{
background:#333;
color:white;
padding:20px;
text-align:center;
}
```

```
nav{
background:#555;
display:flex;
justify-content:center;
gap:20px;
padding:15px;
}
```

```
nav a{
color:white;
```

```
text-decoration:none;
}
```

```
.container{
display:flex;
padding:20px;
gap:20px;
}
```

```
.content{
flex:3;
background:white;
padding:20px;
}
```

```
.sidebar{
flex:1;
background:#ddd;
padding:20px;
}
```

```
footer{
background:#333;
color:white;
padding:20px;
text-align:center;
margin-top:20px;
}
```

```
@media(max-width:768px){
```

```
nav{
flex-direction:column;
align-items:center;
}
```

```
.container{
flex-direction:column;
}
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Responsive Retail Website</h1>
```

</header>

<nav>

Home

Products

Offers

Contact

</nav>

<div class="container">

<div class="content">

<h2>Welcome</h2>

<p>This webpage automatically adjusts for desktop, tablet and mobile devices using CSS Flexbox and Media Queries.</p>

</div>

<div class="sidebar">

<h3>Latest Offers</h3>

<p>Discounts up to 50%</p>

</div>

</div>

<footer>

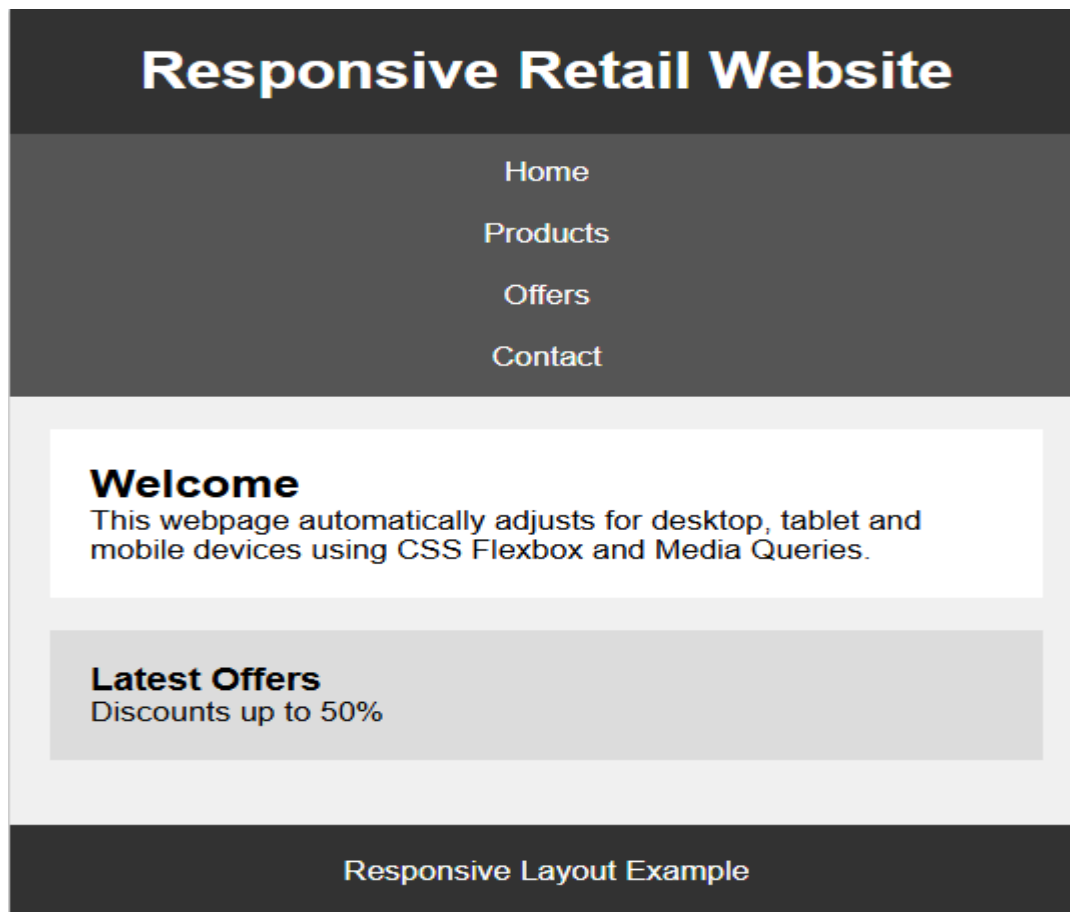
Responsive Layout Example

</footer>

</body>

</html>

OUTPUT:



Experiment 2: Form Validation using JavaScript

CODE:

```
<!-- Experiment 2 - Form Validation -->
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Registration Form</title>
```

```
<style>
```

```
body{  
font-family:Arial;  
background:#f2f2f2;  
}
```

```
.container{
```

```
width:350px;
margin:40px auto;
background:white;
padding:20px;
border-radius:10px;
}
```

```
input{
width:100%;
padding:10px;
margin-top:10px;
margin-bottom:5px;
}
```

```
button{
width:100%;
padding:10px;
margin-top:10px;
}
```

```
.error{
color:red;
font-size:14px;
}
```

```
.success{
color:green;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Registration Form</h2>
```

```
<form onsubmit="return validate()">
```

```
<input type="email" id="email" placeholder="Enter Email">
```

```
<div id="emailError" class="error"></div>
```

```
<input type="tel" id="phone" placeholder="Enter Phone Number">
```

```
<div id="phoneError" class="error"></div>
```

```
<button type="submit">Register</button>
```

```
</form>

</div>

<script>

function validate(){

let email=document.getElementById("email").value;

let phone=document.getElementById("phone").value;

let emailPattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;

let phonePattern=/^[0-9]{10}$/;

document.getElementById("emailError").innerHTML="";

document.getElementById("phoneError").innerHTML="";

let valid=true;

if(!emailPattern.test(email)){
document.getElementById("emailError").innerHTML="Invalid Email";
valid=false;
}

if(!phonePattern.test(phone)){
document.getElementById("phoneError").innerHTML="Phone Number must contain
10 digits";
valid=false;
}

if(valid){
alert("Registration Successful");
}

return valid;

}

</script>

</body>

</html>
```

OUTPUT:

Registration Form

deepak@gmail.com

1334567890

Register

Experiment 3: CSS Layout Debugging

CODE:

```
<!-- Experiment 3 - CSS Layout Debugging -->
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>CSS Layout Debugging</title>
```

```
<style>
```

```
*{  
margin:0;  
padding:0;  
box-sizing:border-box;  
}
```

```
body{  
font-family:Arial;  
background:#f5f5f5;  
}
```

```
.container{  
display:flex;  
flex-wrap:wrap;  
gap:20px;  
padding:20px;  
}
```

```
.box{  
flex:1 1 300px;  
background:#4CAF50;  
color:white;  
padding:30px;
```

```
text-align:center;
font-size:20px;
}
```

```
@media(max-width:768px){
```

```
.container{
flex-direction:column;
}
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1 style="text-align:center;padding:20px;">
CSS Layout Debugging
</h1>
```

```
<div class="container">
```

```
<div class="box">
Content Box 1
</div>
```

```
<div class="box">
Content Box 2
</div>
```

```
<div class="box">
Content Box 3
</div>
```

```
</div>
```

```
</body>
```

</html>

PROGRAM:

CSS Layout Debugging

Content Box 1

Content Box 2